

IS470 Research proposal sketch

Explicit Recommendation incorporating uncertainty and abstention

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1 Introduction and research area

Recommending things is easy, recommending things we think people will like is hard, recommending things people will like *and knowing when we cannot recommend* is harder still. Recommendation research roughly falls into two broad categories, *implicit feedback* and *explicit feedback*. Implicit feedback deals with whether a user will interact with some item in the future, while explicit feedback deals with whether a user will like a particular item, conditionally given that they will interact with the item. Take movie recommendation as an example. An implicit-feedback system tries to answer the question: “will this user watch this movie?” while an explicit-feedback system asks “if the user watches this movie, what rating will they give it?”.

While these two are highly linked, they are evaluated differently because their predictions have different meaning. Implicit recommendation systems usually output a ranking of items for the user to interact with. This means that their performance depends on whether the items the user actually interacts with are placed at the top of the ranking.

Explicit systems take interaction as given. Since we now propose a numerical rating instead of a list of items, we can use metrics such as the Root Mean Square Error (RMSE). Explicit recommendation is often formulated as a matrix completion problem, made famous by the algorithm that won the Netflix recommendation challenge [1]; or a graphical one, popularised by methods such as IGMCM [2]. In both of these structures, they often do not natively encode uncertainty. *Id est*, there is no easy way to say if one numeric rating is made with more gumption than another numeric rating.

Therefore, we extend the above question for explicit-feedback systems and ask “if the user watches this movie, how confident are we that this is the rating will they give the movie? and if we are not confident, can we hold off on making a recommendation?”

This research project aims to improve explicit recommendation systems through joint training and incorporation of recommendation uncertainty with abstention. We enable the system to achieve better performance by having a mathematical structure for both the prediction of how much a user will like an item and for how unsure the model is of its own prediction.

2 Research goal

We want to test if an uncertainty-aware explicit-feedback recommender can be trained, and if abstaining on unreliable predictions yields a better accuracy–coverage trade-off than abstention heuristics that do not use a learned uncertainty.

2.1 Hypothesis

A jointly-trained uncertainty model should outperform naive abstention baselines over the range of budgets we test. Ideally, at zero abstention, the uncertainty-trained model should still beat plain matrix factorisation on full test RMSE.

A priori, understanding the system’s uncertainty about its own recommendation is useful. Therefore, we want to find some mathematical way to understand this uncertainty’s structure and we *hypothesise that this is a low-rank matrix*. The rationale, loosely, is as follows: one of the interpretations for the rating matrix’s low-rank structure is that, across many people, we are not as idiosyncratic as we think. Therefore, everyone’s wants can be explained by a few latent factors. We can then be relatively confident when we recommend something for someone who exhibits behaviour that is aligned to the factors, but are unable to confidently recommend to someone else who exhibits behaviour outside of the latent factors. A fuzzy example is if Timmy is a hardcore Star Wars fan, we can be rather confident of their likes and dislikes based on other Star Wars fans, but we are less confident of Bob, who might exhibit behaviours (like not being that much of a fan for anything) that cannot be inferred from other movie watcher’s habits.

2.2 Motivation

We believe that with uncertainty, it is useful for the research question in two ways. If the entries the model flags as uncertain really are the ones it predicts badly, then withholding a recommendation on these top-ranked fraction removes exactly the predictions that contribute most to the error. The second one is the ability to discount noisy ratings. Some users rate consistently, some do not, and some items are rather controversial. Learning the uncertainty alongside the ratings lets the model discount the entries it cannot fit, so a handful of noisy ratings can no longer distort the model’s predictions. Previous results in [3] show this empirically.

3 Formal problem statement

Let $u_1, \dots, u_n$ be users, indexed by $i \in [n]$, and $v_1, \dots, v_m$ items, indexed by $j \in [m]$. Ratings take values in $\mathcal{R} = \mathbb{R}$. The complete rating matrix $R \in \mathcal{R}^{n \times m}$ is observed only on $\Omega \subseteq [n] \times [m]$ with $|\Omega| \ll nm$; the remaining entries are missing. We split Ω into disjoint $\Omega_{\text{train}}, \Omega_{\text{val}}, \Omega_{\text{test}}$ in proportion 70, 20, 10.

A conventional explicit-feedback recommender is a function

$$f(i, j) = \hat{S}_{ij}, \quad (1)$$

returning a predicted rating given u_i will interact with v_j .

We instead train a model to predict both the rating and its uncertainty.

$$f(i, j) = (\hat{S}_{ij}, \hat{U}_{ij}), \quad (2)$$

where larger $\hat{U}_{ij}$ denotes lower confidence. We require of $\hat{U}$ only that it *order* entries by reliability, not that it be calibrated.

For an abstention rate $p \in [0, 1)$, the model declines to answer on the fraction p of test entries carrying the largest $\hat{U}$, retaining

$$\Omega_p(f) \subseteq \Omega_{\text{test}}, \quad |\Omega_p(f)| = \lceil (1 - p) |\Omega_{\text{test}}| \rceil, \quad (3)$$

and is scored on what remains:

$$\text{RMSE}(f, p) = \sqrt{\frac{1}{|\Omega_p(f)|} \sum_{(i, j) \in \Omega_p(f)} (\hat{S}_{ij} - R_{ij})^2}. \quad (4)$$

At $p = 0$ this reduces to ordinary test RMSE. We would like to make clear that the unit of abstention is per recommendation. This is to say that for the same user, the model might be able to give a rating for one

item but not for another item. Putting it concretely, for each pair (i, j) , declining (i, j) says nothing about (i, j') for a different item j' and vice versa.

4 Current experimental design

4.1 Current model

We will evaluate a naive matrix decomposition recommender system,

$$S = AB^\top, \quad U = CD^\top, \quad (5)$$

with $A \in \mathbb{R}^{n \times k}$, $B \in \mathbb{R}^{m \times k}$ and likewise for C, D . Both are fitted on the observed entries, the only entries at which a residual can be formed:

We co-train both together using the following loss for Ω_{train} :

$$\mathcal{L}(A, B, C, D) = \frac{1}{|\Omega|} \sum_{(i,j) \in \Omega} (S_{ij} - R_{ij})^2 \exp(-U_{ij}) + \lambda \sum_{(i,j) \in \Omega} U_{ij} \quad (6)$$

4.2 Baselines and datasets

A low-rank $\hat{U}$ is only interesting if it abstains better than signals requiring no training. We compare against IGMCM [2], SoftImpute [4], AutoRec [5], plain matrix factorisation [1], and UAIMC [3], which is the nearest prior work and already produces an uncertainty estimate. We also plan to run this on a LightGCN model modified to work for explicit feedback tasks.

Since none of these methods has an abstention mechanism, we will implement three naive abstention mechanisms as a baseline. The first is random abstention (Algorithm 1); the model just randomly decides what not to answer. The second is if the model predicts a low rating for a given test pair (u_i, v_j) , the model can decide not to recommend it and abstain (Algorithm 2). The third is abstention on lower support ratings (Algorithm 3), where, if the model sees that we have very little ratings for the user, the model then chooses not to answer. All three are specified in subsection 4.3.

We additionally compare full test RMSE at $p = 0$ against plain matrix factorisation, matched on k , and hyper-parameters search budget, to test if the uncertainty term improves the rating head even when abstention is disabled. Hyper-parameters for this comparison alone are selected on validation full RMSE, so as not to advantage the uncertainty-equipped model by construction.

We evaluate on the following datasets:

- MovieLens-100K
- MovieLens-1M
- MovieLens-25M
- Douban
- Amazon Games

4.3 Abstention algorithm for baseline models without uncertainty

For this section it is worth outlining, formally, the abstention mechanism algorithm implementation for the models at test time. Since all the models (other than UAIMC) does not produce an uncertainty estimate, we equip each with three abstention rules which do not require uncertainty. We add these abstention systems not as a novel contribution but to ensure that learned uncertainty gets a fair fight.

Algorithm 1 Random abstention

Require: Ω_{test} , budget p

- 1: $b \leftarrow \lfloor p |\Omega_{\text{test}}| \rfloor$
 - 2: $\sigma \leftarrow \text{SHUFFLE}(\Omega_{\text{test}})$
 - 3: **return** $\Omega_p \leftarrow \sigma[b + 1 : |\Omega_{\text{test}}|]$
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Algorithm 2 Low predicted rating abstention

Require: Ω_{test} , model $\hat{S}$, budget p

- 1: $b \leftarrow \lfloor p |\Omega_{\text{test}}| \rfloor$
 - 2: **for all** $(i, j) \in \Omega_{\text{test}}$ **do**
 - 3: $s_{ij} \leftarrow \hat{S}_{ij}$
 - 4: **end for**
 - 5: $\sigma \leftarrow \text{SORTASCENDING}(\Omega_{\text{test}}, \text{key} = s)$
 - 6: **return** $\Omega_p \leftarrow \sigma[b + 1 : |\Omega_{\text{test}}|]$
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Algorithm 3 Low user support abstention

Require: Ω_{test} , Ω_{train} , budget p

- 1: $b \leftarrow \lfloor p |\Omega_{\text{test}}| \rfloor$
 - 2: **for all** users i appearing in Ω_{test} **do**
 - 3: $n_i \leftarrow |\{j : (i, j) \in \Omega_{\text{train}}\}|$ $\triangleright$ support, from training split only
 - 4: $T_i \leftarrow \{(i, j) \in \Omega_{\text{test}}\}$ $\triangleright$ this user's test entries
 - 5: **end for**
 - 6: $\pi \leftarrow \text{SORTASCENDING}(\text{users}, \text{key} = n)$
 - 7: $A \leftarrow \emptyset$ $\triangleright$ abstained set
 - 8: **for all** $i \in \pi$ **do**
 - 9: **if** $|A| + |T_i| \leq b$ **then**
 - 10: $A \leftarrow A \cup T_i$ $\triangleright$ decline this user entirely
 - 11: **else**
 - 12: $A \leftarrow A \cup \text{RANDOMSUBSET}(T_i, b - |A|)$
 - 13: **break**
 - 14: **end if**
 - 15: **end for**
 - 16: **return** $\Omega_p \leftarrow \Omega_{\text{test}} \setminus A$
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Algorithm 3 declines whole users in order of increasing support until the budget is exhausted, so at most one user is split at the boundary.

4.4 Planned experiments

Method (rule)	ML-100K				ML-1M				ML-25M				Douban				Am. Games			
	0	.05	.10	.20	0	.05	.10	.20	0	.05	.10	.20	0	.05	.10	.20	0	.05	.10	.20
Ours ($\hat{U}$)																				
UAIMC ($\hat{U}$)																				
IGMC (rand)																				
IGMC (val)																				
IGMC (supp)																				
SoftImpute (rand)																				
SoftImpute (val)																				
SoftImpute (supp)																				
AutoRec (rand)																				
AutoRec (val)																				
AutoRec (supp)																				
Plain MF (rand)																				
Plain MF (val)																				
Plain MF (supp)																				
LightGCN-e (rand)																				
LightGCN-e (val)																				
LightGCN-e (supp)																				

5 Broader problem space and extensions

The above described experiment is brazenly simple and there are many areas where things could not work for our model and the hypothesis. The biggest one is that, for now, we have abstention per recommendation, whereas an abstention on the user is the more widely deployed idea (cold start soft impute).

An extension could be to replace the main prediction model while retaining the current uncertainty model. For example we can use the low-rank rating predictor while replacing the uncertainty module with a graph-based model, allowing uncertainty to depend on local interaction structure rather than solely on latent user and item factors.

We also need to study if the hypothesis, that U is low rank, holds. If it is not, we can open up our investigation to better understand if there are other structures that are better suited to represent it. In UAIMC for example, the uncertainty comes from a graph structure. The rank of U might be also different to the rank of S . Together, we can further explore how, and why uncertainty arises for explicit recommendation models.

References

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